

Rachid CHEICK MOHAMED

Prospective PhD Candidate in Computer Science / Artificial Intelligence

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About Me

M.Sc. candidate in Artificial Intelligence and Intelligent Systems (AIIS), with a strong background in Deep Learning and Machine Learning for smart systems and cybersecurity applications. Actively working on diverse topics in computer science, AI, and biomedical analysis. In order to achieve my personal goal, I aspire to pursue an academic career as a researcher and university lecturer specialising in Artificial Intelligence.

Research Interests

Deep Learning Architectures • Sequence Modelling & Time-Series Classification • Artificial Intelligence • Network Cyber-Security (NIDS) • Bio-Informatics & Signal Processing • Computer Science

Education

M.Sc. in Artificial Intelligence and Intelligent Systems 2024 – 2026
Gümüşhane University — Gümüşhane, Turkey

- **Thesis:** *Performance Evaluation of Deep Learning Models for Intrusion Detection Using Network Traffic*
- **Supervisor:** Dr. Samet TONYALI
- **Language of Instruction:** Turkish (C1 Academic Proficiency)

B.Sc. in Computer Science 2017 – 2020
University of Ngaoundéré — Cameroon

B.Sc. in Biomedical Analysis 2019 – 2022
ETMS-Yaoundé — Cameroon

Publications & Research Papers

- **Cheick Mohamed Rachid, Samet TONYALI.** “Leak-Free Benchmarking of Deep and Classical Learning Models for Network Intrusion Detection on CIC-IDS2017 and UNSW-NB15.” Submitted to **Elsevier (Engineering Applications of Artificial Intelligence, 2026)** .
[doi:10.13140/RG.2.2.28013.50403](https://doi.org/10.13140/RG.2.2.28013.50403)

Research & Academic Projects

Master’s Thesis Project 2024 – 2026
Gümüşhane University — *Advanced Sequence and Deep Learning Modelling for Network Traffic Analysis*

- Designed and evaluated **1D-CNN** and hybrid **CNN-BiLSTM** architectures against robust baselines (Random Forest, XGBoost) on high-dimensional benchmark datasets (CIC-IDS2017, UNSW-NB15).
- Implemented a rigorous data-engineering pipeline: stratified 5-fold cross-validation with intra-fold SMOTE oversampling to address high class imbalance.
- Applied **Focal Loss** optimisation and **Youden’s J threshold alignment** to improve low-frequency attack pattern recognition in binary and multi-class settings.

- Achieved state-of-the-art results with CNN-BiLSTM, demonstrating superior macro F1-score and AUC-ROC on sequential network traffic data.
- Conducted rigorous statistical validation using the **Wilcoxon signed-rank test** with **Bonferroni correction** to guarantee the significance of cross-validation outputs.
- Supplementary work on homography estimation and fundamental matrix computation for advanced image processing.
- Smart buildings and intelligent systems: IoT architectures for connected building management.

Technical Skills

Programming Languages: Python (Advanced), L^AT_EX (Overleaf), SQL, Bash

Machine / Deep Learning: PyTorch, Scikit-learn, XGBoost, RAPIDS/cuML, imbalanced-learn

Methods & Analytics: Sequence Modelling (BiLSTM), Convolutional Networks (1D-CNN), Time-Series Classification, Non-parametric Statistical Tests

Tools & Ecosystem: Git, Jupyter Notebooks, Google Colab, Linux

Computer Vision: Convolutional Networks

Image Processing: OpenCV

Languages

French Native — academic writing and publication standard

English Fluent / Professional — academic research and publication standard
TOEFL preparation in progress

Turkish Advanced (C1) — full professional and academic fluency throughout graduate studies

References

Dr. Samet TONYALI

Assistant Professor, Department of Software Engineering

Gümüşhane University, Türkiye

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(Thesis Supervisor)